NexABI

Adaptive business intelligence for Nigerian property.

Property sold in instalments the buyer can actually pay, with an intelligence layer over the business that results. It finds what is losing the agency money **without being asked**, says what to do about it, and closes the finding when the condition stops holding. Buyers work out what they can afford. Realtors find out where to sell.

14	₦42.0B	211	0
Detectors sweeping every 15 minutes	Found on the table right now	Automated tests, all passing	Hardcoded insights

What this document covers

01 The business problem

02 The working prototype

03 The solution workflow

04 Tools, platforms and AI models

05 Sample inputs and outputs

06 Evidence of testing

07 Demonstration

08 Expected business impact

Submission · **August 2026** Built with · **PHP, Livewire, MySQL, OpenAI** Status · **Running, seeded, tested**

1 THE BUSINESS PROBLEM

An agency loses money in the places nobody thought to look.

Two problems, and they are one transaction seen from opposite ends. A buyer who cannot pay ₦90m at once, and an agency that cannot see where its own money is leaking.

The buyer is priced out by the shape of the price, not the price

Most people who want property in Lagos are not short of intent — they are short of a way to pay that fits how they earn. **₦90,000,000** ends a conversation. **₦1,500,000 a month over sixty months** starts one. The arithmetic is trivial and almost nobody in the market presents it, so buyers are left doing the division themselves and getting it wrong.

The agency is running on numbers that are already out of date

Nobody notices leads going cold

Qualified leads stop being contacted and no one finds out until somebody happens to run a report — which is exactly the thing nobody thinks to run.

Spend outlives its results

A channel stops converting and keeps drawing budget for months, because the conversion is calculated at quarter end if at all.

Receivables report themselves wrong

A sheet written last Tuesday still calls a three-weeks-late instalment "scheduled". The status was true when it was typed and is a lie now.

Offline spend cannot be judged at all

Billboards, radio and church announcements produce buyers and leave no click. Their return is not merely unknown — it is *uncomputable*.

> **The constraint that shaped every decision:** conventional business intelligence waits to be asked a question, and the conditions that cost the most are precisely the ones nobody thinks to ask about. So this had to look on its own, rank what it found, and **say nothing at all when the business is healthy** — because saying nothing is what makes the times it speaks worth reading.

2 THE FUNCTIONAL PROTOTYPE

Four AI surfaces, four kinds of user, one set of rows.

A running Laravel application — not a mockup and not a demo script. Real authentication, a 26-table relational schema, and an engine that computes everything it displays. Twelve seeded accounts sign in with hashed passwords; role weighting, area isolation and the approval gates all behave as they would in production.

700	4,500	1,917	120
Properties	Leads	Commitments	Realtors

954
Instalments

Roughly **20 months of trading**, generated from a fixed seed so it is byte-identical on every run. The simulator deliberately plants **seven conditions** in the data — stale leads, a burning campaign, a demand gap, payment risk, a complaint cluster, a performance divergence and a funnel erosion. **The detectors are never told about any of them** and rediscover each independently. Delete the failing campaign's rows and its finding disappears on its own.

What the AI actually does, per user

Finds problems `operator`	14 detectors over 9 metric groups, every 15 minutes. Each finding is narrated, ranked, and carries one executable action that changes real rows and logs a before/after diff.
Answers questions `operator`	Natural language over the metrics layer. An intent router first, a model fallback over computed facts second, and an honest refusal third.
Points the realtor `realtor`	Market-wide demand measured against available stock, plus their own conversion by area. Structurally cannot return a colleague's numbers.
Prices the buyer in `buyer`	Affordability computed against the live catalogue — "what fits ₦3m a month" returns the actual units — plus their own balance. Refuses

investment advice.

> **The AI is an accelerator, never a dependency.** With no API key at all the platform runs unchanged: detectors still fire, findings are still ranked, actions still execute, and a deterministic templated narrator writes the prose. An outage degrades the wording, never the product. Every one of the 211 tests runs with the model **disabled**.

3 THE SOLUTION WORKFLOW

From a condition nobody noticed to a row that changed.

1 The business writes rows

Buyers commit, realtors refer, operators approve and record instalments. Nothing is entered twice and there is no export step — the intelligence reads the same rows the business writes.

2 Nine metric groups aggregate, in SQL

Sales, funnel, leads, properties, finance, marketing, customers, satisfaction, agents. Grouping happens in the database and the statistics happen in PHP over the small series that comes back — never by loading rows into memory.

3 Fourteen detectors look, on a schedule

Seven archetypes: threshold, trend, anomaly, cohort, stagnation, funnel, concentration. Each returns zero or more signals, and *zero is a first-class outcome* — that is what separates a detector from a card that is always on screen.

4 Findings are scored and de-duplicated

`priority = severity × confidence × role weight × log(impact)`. Deliberately multiplicative, so a dramatic finding drawn from twelve records never outranks a certain one drawn from four hundred. Rates are Laplace-smoothed and cohorts drop the most recent weeks, because leads captured last week have not had time to convert.

5 The model phrases it — and only phrases it

It receives **facts the metrics layer already computed**, never rows. It is therefore structurally incapable of inventing a number. Narration is cached against a hash of the figures that produced it, so prose is regenerated only when the numbers actually move.

6 The operator presses one thing

Each finding carries the single action worth taking. The action bus mutates the store, measures its own before/after diff, records who did it, and declares which detectors it invalidates so exactly those re-run inside the same request.

7 The finding retires itself

Because the condition no longer holds. **If the action did not actually change anything, the finding stays exactly where it was** — which is the behaviour you want, and the reason none of this can be faked.

Two rules hold the commercial side together, and both are enforced rather than documented. A payment's status is derived from the clock every time it is read, so nothing can sit in the system calling itself scheduled after it fell due. A commission rate is frozen onto the sale at approval, so editing a listing next year cannot rewrite money a realtor already earned.

4 TOOLS, PLATFORMS AND AI MODELS

What it is made of.

AI models — and where deliberately none is used

JOB	MODEL	WHY THIS ONE
Narrating findings	`gpt-4o-mini`	Given computed facts and asked only to phrase them. Runs across every open finding, so the cheap model is the right one.
Open questions	`gpt-4o-mini`	The fallback behind the intent router, over a ~60-figure fact sheet. Must reply `NO_DATA` when the sheet does not cover the question.
Written reports	`gpt-4o-mini`	Executive summaries from scoped headline figures. Cached six hours — the numbers move slower than the page is opened.
Detection & ranking	`no model`	SQL aggregates and hand-written statistics. Free, deterministic, auditable — and the reason a finding can be trusted.
Affordability	`no model`	Arithmetic against the live catalogue. Exact every time, and it returns clickable units rather than a paragraph.
No key configured	`TemplateNarrator`	Grammar-based deterministic prose with the real figures written in. The product is fully functional without any provider.

Programming languages

LANGUAGE	USED FOR	SIZE
PHP 8.3	Engine, metrics, detectors, action bus, platform operations, AI seam	`20,280 lines`
Blade	Four surfaces — public, buyer portal, realtor area, operator console	`10,829 lines`
CSS	One design system, no framework, both themes	`5,052 lines`
JavaScript	Charts, carousel, chrome — plain ES modules, no build step	`2,256 lines`
Tests	17 feature suites	`3,694 lines`

LANGUAGE	USED FOR	SIZE
SQL	26 tables across 9 migrations	—

Platforms and services

Laravel 13.26	Framework, queues, scheduler, transactions, mail
Livewire 4	Server-rendered reactivity — the insight card mutates the database and re-renders itself with no separate API
MySQL 9	Single system of record. The metrics layer speaks `DATE_FORMAT`, `YEARWEEK`, `DATEDIFF`
OpenAI API	Narration, written summaries, the open-question fallback
Resend	Seven transactional flows over HTTP rather than SMTP
PHPUnit 12	211 tests against MySQL, running the real seeder

> **Responsible AI, as three enforced constraints rather than three promises.**
>
> **Facts in, prose out.** Every call receives figures already computed — never raw records, never a customer row. **"I don't know" is representable.** The answer contract returns `null` and the prompt requires a sentinel, because a wrong number costs more than a missing one. **Scoped to the asker.** A realtor's fact sheet carries market aggregates and their own totals and names no colleague; a buyer's carries the catalogue and their own file. All three are tested.

5 SAMPLE INPUTS AND OUTPUTS

Captured from the running system.

Nothing on this page was written by hand. Each is the actual return value of the engine for the input shown, against the seeded dataset.

A. Findings produced with no input at all

The scheduled sweep. These are the twelve currently open — ranked, and each carrying the one action worth taking.

CRITICAL	33 agents are closing at less than half the rate of your top performers → Rebalance 82 leads from the lagging cohort	₦7.6B
CRITICAL	186 of your highest-scoring leads haven't been contacted in a week → Contact all 186 today	₦9.9B
CRITICAL	3-bed Apartment in Lekki is drawing 12.2× normal demand on 4 units of stock → Raise asking price 6% on the 4 remaining units	₦29.5M
CRITICAL	"Meta — Lagos Mainland Push" is converting 8.0× worse than your channel baseline → Pause the campaign	₦16.6M
HIGH	Viewing-to-negotiating conversion has fallen 46 points over 15 weeks → Review site-visit follow-up within 48 hours	—
MEDIUM	185 open leads have gone untouched for 14+ days → Schedule follow-up for all 185	₦17.0B

B. The operator asks a question

INPUT · ROUTED TO THE METRICS LAYER

```
How much is overdue?
```

OUTPUT

₦5.3B is overdue across **41 instalments** and **41 accounts**. **₦3.2B** has aged beyond 90 days. Total outstanding including not-yet-late is **₦5.7B**.

`confidence 0.85` computed, not narrated drills to the rows

INPUT · ROUTER CANNOT PLACE IT, MODEL FALLBACK OVER COMPUTED FACTS

```
Is the business healthy right now?
```

OUTPUT

The business appears to be performing reasonably well, with revenue of **₦14.1B** over the last 90 days and a change of **+1.3%** against the prior 90. However there are concerning factors, such as **₦5.3B** in overdue amounts...

`confidence 0.70` every figure from the fact sheet

INPUT · OUTSIDE THE DATASET

```
What is the airspeed of a swallow?
```

REFUSED

That data is not available. I could not answer that from what this business actually records. Rather than guess at a number — which is the one mistake that would make every other figure here worth doubting — I would rather say so.

`confidence 0.00` offers what it can answer instead

C. The realtor asks where to sell

INPUT

```
Which location should I focus my marketing on?
```

OUTPUT

Lekki is where the pressure is. 188 buyers have asked for Lekki in the last four months against **34 units** on the shelf — roughly **5.5 buyers per unit**. Average stated budget **₦139.5M**, and stock there starts at **₦50,000,000**.

`market-wide demand` measured against available stock

INPUT · PRIVACY BOUNDARY

```
Who is the best realtor in the company?
```

REFUSED BY DESIGN

I don't have the data for that. Demand is market-wide; performance is private. No route returns a colleague's book and the fact sheet carries no other names.

a test fails if any colleague is named

D. The buyer works out what they can afford

INPUT

What can I afford at 3 million a month?

OUTPUT · COMPUTED, NOT NARRATED

120 properties fit ₦3,000,000 a month — across Ajah, Yaba, Chevron and Gbagada. Figures are at the longest term each property offers; a shorter term costs more monthly and hands over sooner.

e.g. **4-Bedroom Terraced Duplex ₦3,000,000**/mo ₦180,000,000 over 60 months

INPUT · OUTSIDE SCOPE

Will Lekki go up in value?

REFUSED BY DESIGN

I don't have the data for that. Nothing in this database knows whether an area is a good investment, and a confident answer would be a financial opinion dressed as a calculation.

6 EVIDENCE OF TESTING

What has actually been checked.

211	1,288	109	0
Automated tests	Assertions	Live audit checks	Known failures

```
$ php artisan test
PASS  211 tests · 1,288 assertions

$ php artisan nexabi:audit
TOTAL 109 checks · 109 ok · 0 failing · 0 missing
```

Two independent harnesses. The suite proves the logic; the audit walks the shipped system and proves it actually contains what the logic describes. Tests run against **MySQL** rather than SQLite — the metrics layer speaks MySQL date functions, and a suite that avoided them would be testing different code. The suite runs **the real seeder** once and rolls each test back, so the fixture and the shipped dataset cannot drift apart.

Speed, measured with the full book loaded

SCREEN	TIME	QUERIES
Public homepage	95 ms	12
Buyer portal	154 ms	15
Realtor dashboard	101 ms	17
Approval queue	216 ms	23
Properties · 700 units	358 ms	15
Operator dashboard	367 ms	31
Insights feed	519 ms	35

Query counts stay flat as the book grows — that is the number that decides whether a page survives a real roll-out. `ScaleTest` fails if nine metric groups exceed 120 queries or a full detector run leaves the request.

What the suites assert

CommitmentTest	Committing holds rather than sells; approval generates the plan and freezes the rate; the schedule sums to the price exactly; **changing a rate after approval does not move earned commission**; one transfer cannot be recorded twice.
ClosedLoopTest	Applying a recommendation moves real rows, is logged with a diff, retires the finding, and cannot be applied twice.
PlantedNarrativesTest	All seven planted conditions are independently rediscovered by detectors that were never told about them.
AdvisorTest	A quoted monthly budget returns only units that genuinely fit it; a realtor's answer names no colleague; a buyer's names no other buyer; both refuse rather than advise.
NarrationTest	Every detector produces prose; a failing model falls through with identical output; **the prompt carries facts and never rows**.
IsolationTest	Operators, realtors and buyers cannot enter each other's areas.
CacheCoherenceTest	The metric cache agrees with the database after an action, a refused action, a rolled-back probe and a clock jump.
AskEngineTest	An unanswerable question comes back saying so, never as a confident wrong number. Every suggested question actually answers.

What has not been proven. Three things, named here rather than left for a judge to find.

The dataset is simulated. 700 properties and 4,500 leads are generated from a fixed seed, not a real agency's book. The engine is proven against a business that behaves plausibly; it has not yet met one that behaves awkwardly.

Detector thresholds are reasoned, not calibrated. Fourteen detectors carry baselines chosen from the data's own distribution. No operator has yet told us which findings were worth acting on, and that feedback is what turns a defensible threshold into a right one.

No independent security review. Area isolation and the tenant scope are covered by tests, and every model call is scoped and asserted — but no adversarial pass has been run against the deployed surface.

7 DEMONSTRATION

Five minutes, four accounts.

```
# about 90 seconds — simulates ~20 months of trading
php artisan migrate:fresh --seed
php artisan nexabi:detect
php artisan serve

# second terminal: detection and the morning digest fire on their own
php artisan schedule:work
```

Twelve accounts, password `password` on every one.

SIGN IN AS	YOU LAND ON
Operator `ada@anchorstone.ng`	One decision, not a wall of charts — plus the AI briefing and the health strip
Realtor `realtor@anchorstone.ng`	Their code, their leads, commission per unit, and where to sell next
Pending realtor `pending@anchorstone.ng`	One holding screen — every other realtor URL redirects back to it
Buyer `client@example.com`	A file that is behind on payments, and their realtor's number on every page

The walkthrough

1 **Open the operator dashboard**

It opens on *one* thing to decide. Twelve findings are live and **₦42.0B** is on the table.

2 **Apply a recommendation**

Press it. Watch the before/after diff appear, the finding close itself, and the entry land in the action log with who did it.

3 **Ask something it cannot answer**

Confirm it declines rather than producing a plausible figure. This is the part worth testing hardest.

4 **Clear the approval queue**

One pending commitment has no realtor attached. The block and its searchable fix are in the same panel.

5 **Switch to the realtor and ask where to sell**

Then ask who the best realtor is, and watch it refuse.

6 **Switch to the buyer**

Ask what they could afford at a different monthly figure, and click straight through to commit.

7 **Advance the Time Machine 30 days**

The business ages, new findings appear, old ones resolve.

> **Applying a recommendation is the part worth watching.** It is the whole claim of the product and the part a slide cannot make — because if the action does not genuinely change the data, the finding does not go away.

8 EXPECTED BUSINESS IMPACT

What changes for an agency that adopts it.

₦42.0B	15 min	60 mo	₦74.7B
Found across 12 open findings	Between sweeps, unprompted	Longest term a buyer can pick	Instalments collected and tracked

Every figure below is what the engine measured on the seeded agency at the moment this was written. They are the **size of the problems found**, not a projection of profit — the claim is that these were invisible before, not that fixing them is free.

For the agency owner	Problems arrive ranked instead of being discovered at quarter end. **₦5.3B** overdue across 41 accounts, **₦3.2B** of it past ninety days — and receivables that cannot report themselves as scheduled after they fall due.
For the sales floor	**186** high-scoring leads untouched for a week, surfaced without anybody running a report, and **33** realtors closing at under half the top cohort's rate — found in a sweep rather than at review time.
For pricing	A segment drawing **12.2×** the portfolio's normal demand against four available units. That is margin left on the table in a market the agency did not know was hot.
For marketing	A campaign converting **8× worse** than baseline, flagged with the spend attached. And billboards, radio and church announcements finally comparable to everything that tracks itself, because cost and expectation are recorded.
For the realtor	Where demand outruns supply, in numbers: **188 buyers** asking for Lekki against **34 units** available. Effort pointed at the segment that converts rather than the one they happen to know.
For the buyer	A ₦180m property becomes **₦3m a month over 60 months** — and they can ask what fits their figure rather than doing the division. Every instalment visible, with status worked out from today's date.

Why the economics work

Detection costs nothing per run. Fourteen detectors over nine metric groups are SQL aggregates and arithmetic — no model is involved in finding a problem, ranking it, or deciding what to do about it. The model is asked only to write the sentence, from figures already computed, and that sentence is cached against a hash of those figures so it is written once and re-used until the numbers move. Affordability, objective ranking and every action handler use no model at all. The AI bill scales with *how often the numbers change*, not with how many people are looking.

> **And the market this opens.** A buyer who cannot write a ₦180m cheque can commit at ₦3m a month. The agency converts a customer it previously could not serve at all, and the platform gains a payment schedule it can measure, chase and forecast. The intelligence layer exists because that business generates exactly the kind of data worth watching — the product and the engine are not two ideas bolted together, they are the same idea from both ends.